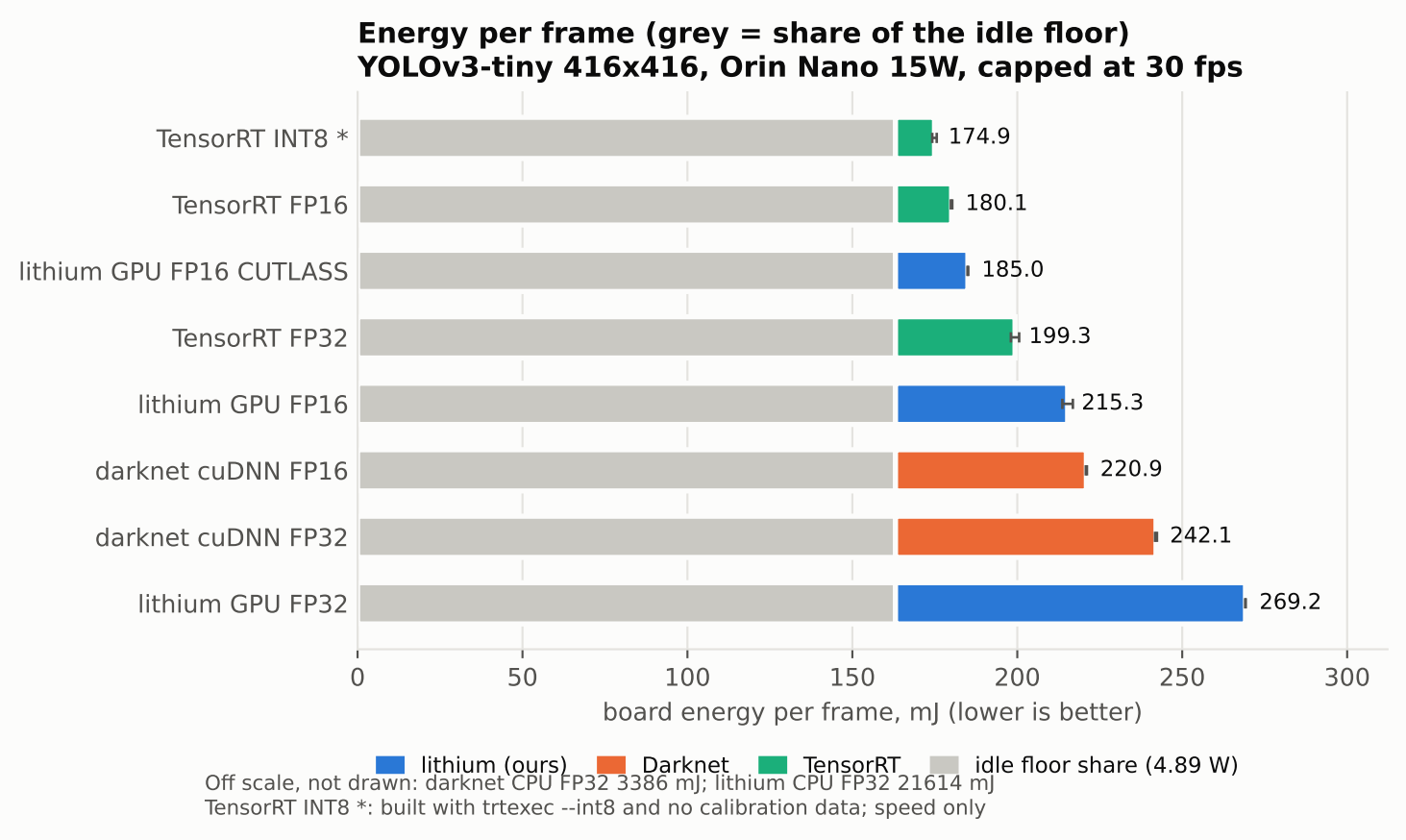
Energy per frame (grey = share of the idle floor)
YOLOv3-tiny 416x416, Orin Nano 15W, capped at 30 fps
TensorRT INT8 *
TensorRT FP16
lithium GPU FP16 CUTLASS
TensorRT FP32
lithium GPU FP16
darknet cuDNN FP16
darknet cuDNN FP32
lithium GPU FP32
174.9
180.1
185.0
199.3
215.3
220.9
242.1
269.2
0
50
100
150
200
250
300
board energy per frame, mJ (lower is better)
lithium (ours)
Darknet
TensorRT
idle floor share (4.89 W)
Off scale, not drawn: darknet CPU FP32 3386 mJ; lithium CPU FP32 21614 mJ
TensorRT INT8 *: built with trtexec --int8 and no calibration data; speed only